

FYS-STK3155/4155 AUG 24



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$$\tilde{y} = X \vec{\Theta}$$

$$X = \begin{bmatrix} x_{00} & x_{01} & \dots & x_{0p-1} \\ x_{10} & & & \\ \vdots & & & \\ x_{n-10} & x_{n-11} & \dots & x_{n-1p-1} \end{bmatrix}$$

polynomial fit

$$\tilde{y}_n = \Theta_0 + \Theta_1 x_n^1 + \Theta_2 x_n^2$$

$$X = \begin{bmatrix} 1 & x_0^1 & x_0^2 \\ 1 & x_1^1 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_{n-1}^1 & x_{n-1}^2 \end{bmatrix}$$

$$\vec{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

$$C(\vec{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$$

MSE

$$\frac{\partial C}{\partial \vec{w}} = \vec{\nabla}_{\vec{w}} C = 0$$

$$\hat{E}_{OLS} = \arg \min_{\vec{w} \in \mathbb{R}^p} \frac{1}{n} \|(y - X\vec{w})\|_2^2$$

$$\hat{E}_{OLS} = \underbrace{(X^T X)^{-1}}_{p \times p} \underbrace{X^T}_{p \times n} \underbrace{y}_n$$

$$\hat{E}_{OLS} \in \mathbb{R}^p$$

$$\tilde{y}_{OLS} = X \hat{\epsilon}_{OLS} = \underbrace{X (X^T X)^{-1} X^T}_{\hat{A}} y$$

$$\hat{A}^2 = \hat{A}$$

idempotent
projection opera-
tor

$$\tilde{y} - \tilde{y}_{OLS} = (\underline{1} - X (X^T X)^{-1} X^T) y$$

y_{OLS} is an optimal model
represented by an orthogo-

nal projection given by
the columns of X

Example

X is an orthogonal
matrix

$$X^T X = X X^T = \underline{\underline{I}}$$

$$\sim$$
$$y_{OLS} = X \underbrace{(X^T X)^{-1}}_{= I} X^T y$$

$$= X X^T \underline{\underline{I}} = y$$

SVD

$$X = U \Sigma V^T$$

$$X \in \mathbb{R}^{n \times p}$$

$$U \in \mathbb{R}^{n \times n}$$

$$U U^T = U^T U = \mathbb{I}$$

$$V \in \mathbb{R}^{p \times p}$$

$$V V^T = V^T V = \mathbb{I}$$

$$\Sigma \in \mathbb{R}^{n \times p}$$

$$\Sigma = \begin{bmatrix} \sigma_0 & & & \\ & \sigma_1 & & \\ & & \ddots & \\ & & & \sigma_{p-1} \\ & & & & 0 \end{bmatrix}$$

$$\sigma_0 > \sigma_1 > \dots > \sigma_{p-1} > 0$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ c & 1 \\ 0 & c \end{bmatrix}$$

$$\Sigma^T \Sigma = \begin{bmatrix} 4 & c \\ 0 & 1 \end{bmatrix}$$

$$\Sigma \Sigma^T = \begin{bmatrix} 4 & c & 0 \\ c & 1 & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\hat{y}_{OLS} = X (X^T X)^{-1} X^T y$$

$$= U \Sigma V^T \underbrace{\left(V \Sigma^T U^T U \Sigma V^T \right)^{-1}}_{\substack{p \times m \quad 1_{m \times m} \quad m \times p}} V \Sigma^T U^T y$$

$$\left(\underbrace{V \Sigma^T \Sigma V^T}_A \underbrace{\quad}_B \right)^{-1} = ?$$

$\checkmark$
 Square and
 invertible

$$B^T = B^{-1}$$

$$(A B)^{-1} = B^{-1} A^{-1}$$

$$\begin{aligned}
 \tilde{y}_{OLS} &= \underbrace{K \Sigma V^T}_{\substack{\uparrow \\ \text{I}}} \underbrace{V}_{\substack{\uparrow \\ \text{I}}} (\Sigma^T \Sigma)^{-1} \\
 &\quad \times \underbrace{V^T V}_{\substack{\uparrow \\ \text{I}}} \Sigma^T K^T y \\
 &= \underbrace{K}_{n \times n} \underbrace{\Sigma}_{n \times p} \underbrace{(\Sigma^T \Sigma)^{-1}}_{p \times p} \underbrace{\Sigma^T}_{p \times n} \underbrace{K^T}_{n \times n} y
 \end{aligned}$$

$$K = 3$$

$$\Sigma = \begin{bmatrix} 2 & c \\ c & 1 \\ c & c \end{bmatrix}$$

$$\Sigma (\Sigma^T \Sigma)^{-1} \Sigma^T = \begin{bmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$U = [\vec{u}_0 \quad \vec{u}_1 \quad \vec{u}_2]$$

$$U \Sigma (\Sigma^T \Sigma)^{-1} \Sigma^T U^T \\ = \vec{u}_0 \vec{u}_0^T + \vec{u}_1 \vec{u}_1^T$$

$$\tilde{y}_{OLS} = \left(\sum_{j=0}^{p-1} u_j u_j^T \right) y$$

$$= \underbrace{u u^T}_{\mathbb{I}} y$$

Ridge

$$\tilde{y}_{\text{Ridge}} = X (X^T X + \lambda \mathbb{I})^{-1} X^T y$$

$\lambda \geq 0$

$$\hat{y}_{\text{Ridge}} = \left(\sum_{j=0}^{p-1} \frac{\sigma_j^2}{\sigma_j^2 + \lambda} u_j u_j^T \right) y$$

Scaling data:

centering and leaving
out intercept.

$$C(\vec{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} \left(y_i - \theta_0 - \sum_{j=1}^{p-1} x_{ij} \theta_j \right)^2$$

$$\nabla_{\theta} C = 0$$

$$\frac{\partial C}{\partial \theta_0} = 0 = -\frac{2}{n} \sum_{i=0}^{n-1} \left(y_i - \theta_0 - \sum_{j=1}^{p-1} x_{ij} \theta_j \right)$$

multiply away $-2/n$

$$\sum_{i=0}^{n-1} \theta_0 = n \theta_0 = \sum_{i=0}^{n-1} y_i - \sum_{i=0}^{n-1} \sum_{j=1}^{p-1} x_{ij} \theta_j$$

simple case $\theta_0 \wedge \theta_1$

$$n E_0 = \sum_{i=0}^{n-1} y_i' - \sum_{i=0}^{n-1} x_{i+1}' E_1$$

$$E_0 = \underbrace{\frac{1}{n} \sum_{i=0}^{n-1} y_i'}_{\mu_y} - E_1 \underbrace{\frac{1}{n} \sum_{i=0}^{n-1} x_{i+1}'}_{\mu_{x_1}}$$

$$\begin{bmatrix} 1 & x_0 \\ 1 & x_1 \\ \vdots & \vdots \\ 1 & x_{n-1} \end{bmatrix}$$

$$\Theta_0 = \mu_y - \sum_{j=1}^{p-1} \mu_{x_j} \Theta_j$$